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Active pen

Abstract

Provided is an active pen including a haptic element, a pressure sensor that, in operation, detects pressure applied to a pen tip, a pen tip electrode provided on the pen tip, a communication device that, in operation, performs wireless communication, and an integrated circuit that, in operation, transmits, from the pen tip electrode, a downlink signal including a pen pressure value indicating a magnitude of the pressure detected by the pressure sensor, in which the integrated circuit, in operation, starts vibrating the haptic element when the communication device receives a control signal that instructs starting of vibration of the haptic element, and stops the vibrating of the haptic element when the pen pressure value falls below a first predetermined value.

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Background/Summary

BACKGROUND

Technical Field

(1) The present disclosure relates to an active pen, and particularly, to an active pen including a haptic element that provides sensory feedback to a user.

Description of the Related Art

(2) There is known a technique in which an electronic apparatus, such as a tablet terminal, controls a haptic element included in an electronic pen, to reproduce the feel of writing on paper. Examples of this type of technique are disclosed in Japanese Patent Laid-Open No. 2019-066960, Japanese Patent Laid-Open No. 2009-238081, and Japanese Patent Laid-Open No. 2002-259044.

(3) There is some time lag in the control when the electronic apparatus controls the haptic element in the electronic pen. Hence, the vibration of the haptic element does not stop immediately after the pen tip has been separated from the panel surface, and the user may feel uncomfortable.

BRIEF SUMMARY

(4) Accordingly, embodiments of the present disclosure provide an active pen that is capable of promptly stopping the vibration of the haptic element after the pen tip has been separated from the panel surface.

(5) A first aspect of the present disclosure provides an active pen including a haptic element, a pressure sensor that, in operation, detects pressure applied to a pen tip, a pen tip electrode provided on the pen tip, a communication device that, in operation, performs near field communication, and an integrated circuit that, in operation, transmits, from the pen tip electrode, a downlink signal including a pen pressure value indicating a magnitude of the pressure detected by the pressure sensor, in which the integrated circuit, in operation, starts vibrating the haptic element when the communication device receives a control signal that instructs starting of vibration of the haptic element, and stops the vibrating of the haptic element when the pen pressure value falls below a first predetermined value.

(6) A second aspect of the present disclosure provides an active pen including a haptic element, a pressure sensor that, in operation, detects pressure applied to a pen tip, and integrated circuit that, in operation, starts vibrating the haptic element in response to a pen-down event, and stops the vibrating of the haptic element in response to a pen-up event, in which the integrated circuit, in operation, stops the vibrating of the haptic element when a pen pressure value indicating a magnitude of the pressure detected by the pressure sensor falls below a first predetermined value.

(7) According to the present disclosure, the vibration of the haptic element can be stopped according to the pen pressure value in the active pen. Therefore, the vibration of the haptic element can be stopped promptly after the pen tip has been separated from the panel surface.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) FIG. 1 depicts a configuration of a position detection system **1** according to an embodiment of the present disclosure;

(2) FIG. 2 depicts a sequence of a process executed in the position detection system **1** according to the embodiment of the present disclosure; and

(3) FIG. 3 depicts a sequence of a process executed in the position detection system **1** according to a modification of the present embodiment.

DETAILED DESCRIPTION

(4) A preferred embodiment of the present disclosure will now be described in detail with reference to the attached drawings.

(5) FIG. 1 depicts a configuration of a position detection system **1** according to the preferred embodiment of the present disclosure. As illustrated in FIG. 1, the position detection system **1** includes an active pen **2** and an electronic apparatus **3** that is a position detection apparatus configured to detect the active pen **2**.

(6) The electronic apparatus **3** is a computer, such as a tablet computer and a digitizer, including a panel surface **3a**. A sensor **30** arranged in the panel surface **3a**, a sensor controller **31** connected to the sensor **30**, a display **32** placed in an overlapping relation with the sensor **30**, a wireless communication device **33**, and a host processor **34** that controls the components of the electronic apparatus **3** including them are provided in the electronic apparatus **3**.

(7) The host processor **34** is a central processing unit of the electronic apparatus **3**, and is configured to read various programs from an unillustrated memory and execute the programs. The

programs executed in this way include various applications including an operating system of the electronic apparatus **3** and a drawing application. Of these applications, the drawing application is a program for executing a process of generating digital ink in reference to the position and the data supplied from the sensor controller **31** and storing the digital ink in the memory of the electronic apparatus **3** and a process of rendering the generated digital ink to generate a video signal indicating the result and supplying the video signal to the display **32**. The display **32** is a device that displays the video signal supplied from the host processor **34**, and includes, for example, a liquid crystal display or an organic electroluminescence (EL) display.

(8) The sensor **30** is a device in which a plurality of sensor electrodes are arranged in the panel surface **3a**. Some of the plurality of sensor electrodes included in the sensor **30** can also be used as common electrodes of the display **32** (electrodes for supplying in common the ground potential to the pixels). When the sensor electrodes are also used as common electrodes, the electronic apparatus **3** provides what is generally called an “in-cell” position detection apparatus. On the other hand, when the sensor electrodes are not used as common electrodes, the electronic apparatus **3** provides what is generally called an “on-cell” or “out-cell” position detection apparatus. The present disclosure can suitably be applied to any of the electronic apparatuses **3**.

(9) The sensor controller **31** is an integrated circuit with a function of detecting the position of the active pen **2** in the panel surface **3a**. Although an active electrostatic system (AES) is used to execute the detection in the example described in the present embodiment, the present disclosure can also suitably be applied when another system, such as an electromagnetic resonance system (EMR), is used to detect the position of the active pen **2**.

(10) The sensor controller **31** corresponding to the AES is configured to perform two-way communication with the active pen **2** through capacitive coupling between the sensor **30** and an electrode (pen tip electrode **21** described later) in the active pen **2**. Hereinafter, a signal transmitted from the sensor controller **31** to the active pen **2** through the communication will be referred to as an “uplink signal US,” and a signal transmitted from the active pen **2** to the sensor controller **31** will be referred to as a “downlink signal DS.”

(11) The uplink signal US is a signal periodically transmitted by the sensor controller **31**. The uplink signal US plays a role of notifying the active pen **2** of the transmission timing of the downlink signal DS and the reception timing of the next uplink signal US and plays a role of transmitting a command to the active pen **2**. On the other hand, the downlink signal DS is a signal including a position signal, which is an unmodulated carrier signal, and a data signal, which is a carrier signal modulated by data. The position signal is a signal for causing the sensor controller **31** to detect the position of the active pen **2**, and the sensor controller **31** is configured to derive the position of the active pen **2** in the panel surface **3a** in reference to the distribution of the reception strength of the position signal in each sensor electrode of the sensor **30**. The data transmitted by the data signal includes switch information and a pen pressure value described later, as well as data for which transmission is instructed by a command. The sensor controller **31** demodulates the data signal to acquire the data transmitted by the active pen **2**. The position derived by the sensor controller **31** and the data acquired by the sensor controller **31** in this way are sequentially supplied from the sensor controller **31** to the host processor **34**.

(12) The wireless communication device **33** is a communication device for performing wireless, near field communication as exemplified by Bluetooth (registered trademark), and is connected to the host processor **34**. The host processor **34** performs near field communication with the active pen **2** and other devices (such as a keyboard, a mouse, a speaker, and headphones) through the wireless communication device **33**.

(13) The signal transmitted by the host processor **34** to the active pen **2** through the wireless communication device **33** includes a control signal of a haptic element **26** described later. The host processor **34** is configured to detect a pen-down event and transmit a control signal that instructs the haptic element **26** to start vibrating, when the pen pressure value supplied from the sensor

controller **31** exceeds a predetermined value V2 (typically, $V2=0$). On the other hand, the host processor **34** is configured to detect a pen-up event and transmit a control signal that instructs the haptic element **26** to stop vibrating, when the pen pressure value supplied from the sensor controller **31** reaches the predetermined value V2. This will be described in more detail later with reference to FIG. 2.

(14) The active pen **2** is an electronic pen (stylus) corresponding to the AES, and includes a core body **20**, the pen tip electrode **21**, a side switch **22**, a pressure sensor **23**, a battery **24**, a wireless communication device **25**, the haptic element **26**, and an integrated circuit **27** as illustrated in FIG. 1. The core body **20** is a member that configures a pen axis of the active pen **2**. The front end of the core body **20** configures the pen tip of the active pen **2**, and the back end is abutted to the pressure sensor **23**. The pen tip electrode **21** is a conductor arranged on the pen tip, and is electrically connected to the integrated circuit **27**. Note that a plurality of pen tip electrodes **21** may be provided in the active pen **2**, and, for example, different pen tip electrodes **21** may be used for transmission and reception.

(15) The side switch **22** is a press-button switch provided on the surface of the active pen **2**, and the user can turn on and off the side switch **22**. The operation state (on-off state) of the side switch **22** is supplied as, for example, switch information of 2 bits to the integrated circuit **27**. The pressure sensor **23** is a sensor that detects the pressure applied to the front end (pen tip) of the core body **20**. The pressure detected by the pressure sensor **23** is supplied as, for example, a pen pressure value of 12 bits to the integrated circuit **27**. The battery **24** plays a role of supplying power necessary for the integrated circuit **27** to operate.

(16) The wireless communication device **25** is a communication device for performing wireless, near field communication as exemplified by Bluetooth (registered trademark). The integrated circuit **27** performs near field communication with the host processor **34** of the electronic apparatus **3** through the wireless communication device **25**. The signal received by the integrated circuit **27** from the host processor **34** through the near field communication includes a control signal of the haptic element **26**.

(17) The haptic element **26** is a device that provides sensory feedback to the user of the active pen **2**, and, for example, includes a vibrator of an eccentric rotating mass system, a linear resonant actuator system, or a piezo (piezoelectric) actuator system. A magnetic fluid in which the hardness can be controlled by the frequency of the applied pulse current may be used to provide the haptic element **26**. The following description is based on the assumption that the haptic element **26** includes the vibrator of the eccentric rotating mass system. In this case, the haptic element **26** includes a motor and an eccentric weight attached to the axis of rotation of the motor.

(18) The integrated circuit **27** is a circuit that executes various processes for communicating with the sensor controller **31** through the pen tip electrode **21** and communicating with the host processor **34** through the wireless communication device **25**. Specifically, the integrated circuit **27** executes a process of detecting a change in the potential of the pen tip electrode **21** to receive the uplink signal US, a process of using the reception timing of the uplink signal US as reference time to determine a transmission and reception schedule of the downlink signal DS and the next uplink signal US, a process of generating the downlink signal DS according to the command included in the uplink signal US, and a process of changing the potential of the pen tip electrode **21** according to the determined transmission and reception schedule to transmit the generated downlink signal DS. The integrated circuit **27** also pairs with the host processor **34** through the wireless communication device **25**, receives various control signals from the paired host processor **34**, and executes processes based on the received control signals.

(19) The processes based on the received control signals include a process of starting or stopping the vibration of the haptic element **26**. Specifically, the integrated circuit **27** is configured to start or stop the vibration of the haptic element **26** according to the control signal transmitted by the host processor **34**. Here, the integrated circuit **27** also executes the process of stopping the vibration of

the haptic element **26** when the pen pressure value supplied from the pressure sensor **23** reaches a value equal to or smaller than a predetermined value **V1**. The predetermined value **V1** is a value ($V1 > V2$) larger than the predetermined value **V2** (value used by the host processor **34** to detect pen-down and pen-up). In the actual process, the integrated circuit **27** stops the vibration of the haptic element **26** at the timing that the wireless communication device **25** receives the control signal that instructs the haptic element **26** to stop vibrating or at the timing that the pen pressure value supplied from the pressure sensor **23** falls below the predetermined value **V1**, whichever comes first.

(20) FIG. **2** depicts a sequence of the process executed in the position detection system **1**. Only the process related to the control of the haptic element **26** is extracted and illustrated in FIG. **2**. The process executed by the active pen **2**, the sensor controller **31**, and the host processor **34** in relation to the control of the haptic element **26** will be described below with reference to FIG. **2**.

(21) First, as illustrated in FIG. **2**, once the active pen **2** transmits the downlink signal **DS** (**S1**), the sensor controller **31** derives the coordinates indicating the position of the active pen **2** in the panel surface **3a** in reference to the position signal in the downlink signal **DS**, acquires the pen pressure value and other data from the data signal in the downlink signal **DS**, and supplies the data to the host processor **34** (**S2**). The process of **S1** and **S2** is executed every time the active pen **2** transmits the downlink signal **DS**.

(22) Next, every time the pen pressure value is supplied from the sensor controller **31**, the host processor **34** determines whether or not there is occurrence of pen-down or pen-up in reference to the series of pen pressure values supplied so far (**S3** and **S4**) and uses the near field communication to transmit the control signal of the haptic element **26** according to the result of the determination (**S5** and **S6**). Specifically, the host processor **34** detects occurrence of pen-down when the pen pressure value changes from a value equal to or smaller than the predetermined value **V2** (typically, $V2=0$) to a value larger than the predetermined value **V2** (**S3**) and transmits a control signal that instructs the haptic element **26** to start vibrating (**S5**). Further, the host processor **34** detects occurrence of pen-up when the pen pressure value changes from a value larger than the predetermined value **V2** to a value equal to or smaller than the predetermined value **V2** (**S4**) and transmits a control signal that instructs the haptic element **26** to stop vibrating (**S6**).

(23) After receiving the control signal that instructs the haptic element **26** to start vibrating, the integrated circuit **27** of the active pen **2** starts vibrating the haptic element **26** (**S7**). Specifically, the integrated circuit **27** applies a current in a first direction to the motor included in the haptic element **26**, to start vibrating the haptic element **26**. As a result, when the user slides the active pen **2** on the panel surface **3a**, the user can feel vibration similar to the vibration in sliding a real writing tool, such as a ballpoint pen, on paper.

(24) While vibrating the haptic element **26**, the integrated circuit **27** executes a process of comparing the supplied pen pressure value and the predetermined value **V1** every time the pen pressure value is supplied from the pressure sensor **23** (**S8**). The integrated circuit **27** stops vibrating the haptic element **26** when detecting that the pen pressure value has become a value equal to or smaller than the predetermined value **V1** as a result of the comparison (**S9**). The integrated circuit **27** also stops vibrating the haptic element **26** when receiving the control signal that instructs the haptic element **26** to stop vibrating (**S10**). Only one of **S9** and **S10** that is executed earlier is actually executed.

(25) Here, in stopping the vibration of the haptic element **26** at **S9** or **S10**, it is preferable that the integrated circuit **27** apply a current in a direction opposite the first direction to the motor and then stop applying the current to the motor to stop the vibration of the haptic element **26**. This can stop the residual vibration of the haptic element **26** caused by inertia more promptly than in a case in which the application of the current to the motor is stopped without the current being applied in the direction opposite the first direction to the motor.

(26) As described above, according to the position detection system **1** of the present embodiment,

the vibration of the haptic element **26** can be stopped according to the pen pressure value in the active pen **2**. This can stop the vibration of the haptic element **26** promptly after the pen tip has been separated from the panel surface **3a**.

(27) Further, according to the active pen **2** of the present embodiment, the vibration of the haptic element **26** is stopped when the pen pressure value falls below the predetermined value V1 that is larger than the predetermined value V2 generally used for the determination of pen-down and pen-up. Hence, even if there is residual vibration of the haptic element **26** caused by inertia, the vibration of the haptic element **26** can be stopped immediately after the pen tip has been separated from the panel surface **3a**.

(28) Note that the present disclosure can also suitably be applied when the predetermined value V1 is equal to the predetermined value V2. In this case, the residual vibration of the haptic element **26** caused by inertia may remain after the pen tip has been separated from the panel surface **3a**. However, the vibration of the haptic element **26** can be stopped more promptly than in the control from the electronic apparatus **3**.

(29) Further, according to the active pen **2** of the present embodiment, the current in the direction opposite the previous direction is applied to the motor in stopping the vibration of the haptic element **26**. This can promptly stop the residual vibration of the haptic element **26** caused by inertia.

(30) Although the preferred embodiment of the present disclosure has been described, the present disclosure is not limited to the embodiment in any way, and it is obvious that the present disclosure can be carried out in various modes without departing from the scope of the present disclosure.

(31) FIG. **3** depicts a sequence of the process executed in the position detection system **1** according to a modification of the present embodiment. The present modification is different from the present embodiment in that the process of **S10** to **S13** is executed in place of **S3** to **S6** illustrated in FIG. **2**.

(32) More specifically, the sensor controller **31** in the present modification determines whether or not there is occurrence of pen-down or pen-up in reference to the series of pen pressure values acquired so far (**S10** and **S11**) and uses the uplink signal US to transmit the control signal of the haptic element **26** according to the result of the determination (**S12** and **S13**). The specific process of **S10** to **S13** is similar to that of **S3** to **S6** except that the uplink signal US is used instead of the near field communication.

(33) In this way, according to the present disclosure, the vibration of the haptic element **26** can also be stopped according to the pen pressure value in the active pen **2** even when the sensor controller **31** uses the uplink signal US to transmit the control signal of the haptic element **26**. This can stop the vibration of the haptic element **26** promptly after the pen tip has been separated from the panel surface **3a**.

(34) The various embodiments described above can be combined to provide further embodiments. All of the U.S. patents, U.S. patent application publications, U.S. patent applications, foreign patents, foreign patent applications and non-patent publications referred to in this specification and/or listed in the Application Data Sheet are incorporated herein by reference, in their entirety. Aspects of the embodiments can be modified, if necessary to employ concepts of the various patents, applications and publications to provide yet further embodiments.

(35) These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

Claims

1. An active pen, comprising: a haptic element; a pressure sensor that, in operation, detects pressure applied to a pen tip; a pen tip electrode provided on the pen tip; a communication device that, in operation, performs wireless communications; and an integrated circuit that, in operation, transmits, from the pen tip electrode, a downlink signal including a pen pressure value indicating a magnitude of the pressure detected by the pressure sensor, wherein the integrated circuit, in operation, starts vibrating the haptic element when the communication device receives, from a device external to the active pen, a control signal that instructs starting of vibration of the haptic element, and stops the vibrating of the haptic element when the pen pressure value falls below a first predetermined value independently from a status of the control signal, wherein the haptic element includes a motor, and wherein the integrated circuit, in operation, applies a first current in a first direction to the motor to start the vibrating of the haptic element, and applies a second current in a direction opposite the first direction to the motor and then stops applying the second current to the motor to stop the vibrating of the haptic element when the pen pressure value falls below the first predetermined value.
2. The active pen according to claim 1, wherein: the device external to the active pen is a host processor connected to a sensor controller that, in operation, receives the downlink signal through capacitive coupling between a sensor arranged in a panel surface and the pen tip electrode, and the host processor, in operation, wirelessly transmits the control signal that instructs starting of the vibration of the haptic element when the pen pressure value supplied from the sensor controller exceeds a second predetermined value.
3. An active pen, comprising: a haptic element; a pressure sensor that, in operation, detects pressure applied to a pen tip; a pen tip electrode provided on the pen tip; a communication device that, in operation, performs wireless communications; and an integrated circuit that, in operation, transmits, from the pen tip electrode, a downlink signal including a pen pressure value indicating a magnitude of the pressure detected by the pressure sensor, wherein the integrated circuit, in operation, starts vibrating the haptic element when the communication device receives, from a device external to the active pen, a control signal that instructs starting of vibration of the haptic element, and stops the vibrating of the haptic element when the pen pressure value falls below a first predetermined value independently from a status of the control signal, wherein the device external to the active pen is a host processor connected to a sensor controller that, in operation, receives the downlink signal through capacitive coupling between a sensor arranged in a panel surface and the pen tip electrode, and wherein the host processor, in operation, wirelessly transmits the control signal that instructs starting of the vibration of the haptic element when the pen pressure value supplied from the sensor controller exceeds a second predetermined value, and wherein the first predetermined value is larger than the second predetermined value.
4. The active pen according to claim 2, wherein: the host processor, in operation, wirelessly transmits a control signal that instructs stopping of the vibration of the haptic element when the pen pressure value supplied from the sensor controller falls below the first predetermined value.
5. An active pen comprising: a haptic element; a pressure sensor that, in operation, detects pressure applied to a pen tip; and an integrated circuit that, in operation, starts vibrating the haptic element in response to receiving, from a device external to the active pen, a control signal that instructs starting of vibration of the haptic element, wherein the integrated circuit, in operation, stops the vibrating of the haptic element when a pen pressure value indicating a magnitude of the pressure detected by the pressure sensor falls below a first predetermined value independently from a status of the control signal; a pen tip electrode provided on the pen tip; and a communication device that, in operation, performs wireless communication, wherein: the integrated circuit, in operation, transmits, from the pen tip electrode, a downlink signal including the pen pressure value indicating the magnitude of the pressure detected by the pressure sensor, the device external to the active pen is a host processor connected to a sensor controller that, in operation, receives the downlink signal

through capacitive coupling between a sensor arranged in a panel surface and the pen tip electrode, and the host processor, in operation, detects a pen-down event when the pen pressure value supplied from the sensor controller exceeds a second predetermined value, and wirelessly transmits the control signal that instructs the starting of the vibration of the haptic element to thereby start the vibrating of the haptic element when the pen-down event is detected, and the first predetermined value is larger than the second predetermined value.
